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Force feedback module

Abstract

A force feedback module is provided. The force feedback module includes a trigger element, an actuating element, and a transmission assembly disposed between the trigger element and the actuating element. The transmission assembly includes a first transmission element. The first transmission element and the trigger element change between a contact state and a non-contact state. When the first transmission element and the trigger element are in the contact state, a driving force generated by the actuating element is transmitted to the trigger element via the transmission assembly to generate force feedback.

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Field of Classification Search

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application claims the benefit of Taiwan Patent Application No. 111210472, filed on Sep. 26, 2022, the entirety of which is incorporated by reference herein.

BACKGROUND OF THE INVENTION

Field of the Invention

(2) The present invention relates to a force feedback module.

Description of the Related Art

(3) As technology has improved, virtual reality (VR) and augmented reality (AR) have undergone significant development. Nowadays, there is already a wide selection of smart VR wearable devices and smart AR wearable devices. When a user puts on a smart wearable device, he can see a vivid and realistic display. To further enhance user experience in haptic interactions, a handheld controller may be used. Common handheld controllers include handle-type controllers and gun-type controllers, which may have buttons or triggers. Users may interact with the display by pressing the button or pulling the trigger. However, current handheld controllers do not provide immersive force feedback when a user presses the button or pulls the trigger.

BRIEF SUMMARY OF THE INVENTION

(4) A force feedback module is provided in accordance with some embodiments. The force feedback module includes a trigger element, an actuating element, and a transmission assembly disposed between the trigger element and the actuating element. The transmission assembly includes a first transmission element. The first transmission element and the trigger element change between a contact state and a non-contact state. When the first transmission element and the trigger

element are in the contact state, a driving force generated by the actuating element is transmitted to the trigger element via the transmission assembly to generate force feedback.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 is a perspective view of the force feedback module.

(3) FIG. 2 is an exploded view of the force feedback module.

(4) FIG. 3 is a front view of the force feedback module illustrated in FIG. 1.

(5) FIG. 4 is a top view of the force feedback module illustrated in FIG. 1.

(6) FIG. 5 is a perspective view of the force feedback module, wherein the trigger element is in contact with the first transmission element.

(7) FIG. 6 is a front view of the force feedback module illustrated in FIG. 5.

(8) FIG. 7 is a top view of the force feedback module illustrated in FIG. 5.

(9) FIG. 8 is a perspective view of the force feedback module, wherein the trigger element and the first transmission element rotate to the limit.

(10) FIG. 9 is a front view of the force feedback module illustrated in FIG. 8.

(11) FIG. 10 is a top view of the force feedback module illustrated in FIG. 8.

DETAILED DESCRIPTION OF THE INVENTION

(12) The following disclosure provides many different embodiments, or examples, for implementing different features of the subject matter provided. Specific examples of components and arrangements are described below to simplify this disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature “on” and/or “above” a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, so that the first and second features may not be in direct contact. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations), and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(13) In this specification, the words “including”, “comprising”, “having”, and the like are open words, so they should be interpreted as meaning “including but not limited to . . .”. Therefore, when the words “including”, “comprising”, “having”, and the like are used in the description of this disclosure, the presence of corresponding features, regions, steps, operations and/or components is specified, and without excluding the presence of one or more other features, regions, steps, operations and/or components. In addition, in this specification, words “about” or “substantially” are generally interpreted as within 10%, 5%, 3%, 2%, 1%, or 0.5% of a given value or range.

(14) Please refer to FIG. 1 and FIG. 2 to understand a force feedback module **100**. FIG. 1 is a perspective view of the force feedback module **100**. FIG. 2 is an exploded view of the force feedback module **100**. A handheld controller that includes a button or a trigger may be equipped with the force feedback module **100**. Therefore, force feedback may be provided when a user applies force to (e.g. press) the button or the trigger. In this embodiment, the force feedback module **100** includes a trigger element **200**, a transmission assembly **300**, an actuating element **400**, a circuit element **500**, a sensing element **600**, and a pin **700**. However, elements may be added or omitted according to actual needs.

(15) The trigger element **200** may be disposed close to the fingers of the user. The trigger element **200** includes a trigger body **210** and a trigger pivotally connected portion **220**. The shape of the trigger body **210** may correspond to the shape of the casing of the button or the trigger. A hole **221** may be formed on the trigger pivotally connected portion **220**, so the pin **700** may pass through the trigger pivotally connected portion **220**. The trigger element **200** may rotate around the pin **700**.

(16) The transmission assembly **300** is disposed between the trigger element **200** and the actuating element **400**. The transmission assembly **300** includes a first transmission element **310**, a second transmission element **320**, a third transmission element **330**, a fourth transmission element **340**, and a transmission rod **350**. In some embodiments, the first transmission element **310** may be a linear gear. In some embodiments, the second transmission element **320**, the third transmission element **330**, and the fourth transmission element **340** may be a circular gear. For example, the fourth transmission element **340** may be a bevel gear, but it is not limited thereto. For ease of illustration, the teeth portions of the first transmission element **310**, the second transmission element **320**, the third transmission element **330**, and the fourth transmission element **340** are not illustrated. It should be noted that the portions where the first transmission element **310** and the second transmission element **320** are in contact with each other may have meshing teeth portions. In addition, the portions where the third transmission element **330** and the fourth transmission element **340** are in contact with each other may have meshing teeth portions.

(17) The first transmission element **310** includes a transmission body **311** and a transmission pivotally connected portion **312**. In some embodiments, the transmission body **311** is substantially L-shaped and includes a straight-line portion **3111** and a curved-line portion **3112**. The transmission body **311** may include a protrusion **3113** facing the trigger element **200**. A hole **3121** may be formed on the transmission pivotally connected portion **312**, so the pin **700** may pass through the transmission pivotally connected portion **312**. The hole **3121** of the transmission pivotally connected portion **312** may be aligned with the hole **221** of the trigger pivotally connected portion **220**. The first transmission element **310** may rotate around the pin **700**.

(18) The second transmission element **320** is in contact with the first transmission element **310**. In some embodiments, the second transmission element **320** meshes (or engage) the first transmission element **310**. In some embodiments, the second transmission element **320** is connected to the third transmission element **330** via the transmission rod **350**. That is, the second transmission element **320** and the third transmission element **330** have the same rotation axis. The second transmission element **320** and the third transmission element **330** may rotate around the transmission rod **350** (i.e. the transmission rod **350** is the rotation axis). The fourth transmission element **340** is in contact with the third transmission element **330**. In some embodiments, the fourth transmission element **340** meshes (or engage) the third transmission element **330**.

(19) The actuating element **400** may be any elements that can generate a driving force, and the generated driving force may have different amount and directions. For example, the actuating element **400** may generate the driving force by rotation, translation, vibration, and the like. In some embodiments, the actuating element **400** may be a motor. In some embodiments, the actuating element **400** includes a rotation shaft **410** (denoted in FIG. 3). The fourth transmission element **340** may be connected to the actuating element **400** via the rotation shaft **410**, and the fourth transmission element **340** may rotate around the rotation shaft **410**.

(20) The circuit element **500** is disposed close to the second transmission element **320**. In some embodiments, the circuit element **500** may be a circuit board. For example, the circuit element **500** may be a rigid board, a rigid-flex board, and the like, but it is not limited thereto. The sensing element **600** is disposed on the circuit element **500**. The sensing element **600** may sense the position of the first transmission element **310** and generate a signal. In some embodiments, the transmission rod **350** is embedded (or engaged) in the sensing element **600**, so the sensing element **600** is able to sense the position of the first transmission element **310** via the rotation of the transmission rod **350**, but the sensing of the sensing element **600** is not limited thereto. The pin **700**

passes through the hole **221** of the trigger pivotally connected portion **220** and the hole **3121** of the transmission pivotally connected portion **312**. In some embodiments, the pin **700** is cylindrical-shaped.

(21) As described above, the trigger element **200** and the first transmission element **310** are both able to rotate around the pin **700**. Please refer to FIG. 1 and FIG. 3 to FIG. 7 to understand the rotation of the trigger element **200**. FIG. 3 is a front view of the force feedback module **100** illustrated in FIG. 1. FIG. 4 is a top view of the force feedback module **100** illustrated in FIG. 1. FIG. 5 is a perspective view of the force feedback module **100**, wherein the trigger element **200** is in contact with the first transmission element **310**. FIG. 6 is a front view of the force feedback module **100** illustrated in FIG. 5. FIG. 7 is a top view of the force feedback module **100** illustrated in FIG. 5.

(22) Users may apply force to the button or the trigger in the handheld controller, making the trigger element **200** rotates around the pin **700** toward the first transmission element **310** until the trigger element **200** is in contact with the first transmission element **310** (for example, the trigger element **200** may be in contact with the protrusion **3113** of the first transmission element **310**). In some embodiments, during the period that the trigger element **200** rotates around the pin **700** toward the first transmission element **310** until being in contact with the first transmission element **310**, the first transmission element **310** may not rotate.

(23) Next, please refer to FIG. 5 to FIG. 10 to understand the rotation of the first transmission element **310**. FIG. 8 is a perspective view of the force feedback module **100**, wherein the trigger element **200** and the first transmission element **310** rotate to the limit. FIG. 9 is a front view of the force feedback module **100** illustrated in FIG. 8. FIG. 10 is a top view of the force feedback module **100** illustrated in FIG. 8.

(24) After the trigger element **200** contacts the first transmission element **310** (as illustrated in FIG. 5 to FIG. 7), users may continuously apply force to the button or the trigger in the handheld controller, making the trigger element **200** and the first transmission element **310** rotated around the pin **700** together (direction of rotation is illustrated by the arrow R in FIG. 5) until the trigger element **200** and the first transmission element **310** rotate to the limit (i.e. a specific position) and cannot rotate further (as shown in FIG. 8 to FIG. 10). In some embodiments, when the trigger element **200** and the first transmission element **310** rotate to the limit (i.e. a specific position) and cannot rotate further, the second transmission element **320** contacts the straight-line portion **3111** of the first transmission element **310** so as to stop the rotation of the trigger element **200** and the first transmission element **310**, but stopping the rotation of the trigger element **200** and the first transmission element **310** is not limited thereto.

(25) In the embodiments illustrated in FIG. 1, FIG. 3, and FIG. 4, the trigger element **200** and the first transmission element **310** may be referred to as in a non-contact state, and in the embodiments illustrated in FIG. 5 to FIG. 10, the trigger element **200** and the first transmission element **310** may be referred to as in a contact state. In other words, the trigger element **200** and the first transmission element **310** are able to alternate between the non-contact state and the contact state.

(26) When the trigger element **200** and the first transmission element **310** are in the non-contact state, a gap is formed between the trigger element **200** and the first transmission element **310** (such gap may be referred to as “an idle stroke”), so the driving force generated by the actuating element **400** cannot be transferred to the trigger element **200** via the transmission assembly **300**. That is, when the trigger element **200** and the first transmission element **310** are in the non-contact state, applying force to the button or the trigger of the handheld controller would not generate force feedback.

(27) When the trigger element **200** and the first transmission element **310** are in the contact state and the actuating element **400** generates the driving force, the actuating element **400** drives the fourth transmission element **340** to rotate around the rotation shaft **410** of the actuating element **400**. The fourth transmission element **340** may drive the third transmission element **330** and the

second transmission element **320** to rotate with the transmission rod **350** as the rotating shaft. Furthermore, the second transmission element **320** may drive the first transmission element **310** to rotate with the pin **700** as the fulcrum. Therefore, the driving force generated by the actuating element **400** can adjust the position of the first transmission element **310** and then transferred to the trigger element **200**. That is, when the trigger element **200** and the first transmission element **310** are in the contact state and the actuating element **400** generates the driving force, the transmission assembly **300** is able to transmit the driving force generated by the actuating element **400** to the trigger element **200** so as to generate force feedback.

(28) Due to the rotation, translation, or vibration of the actuating element **400**, users' fingers may experience different feelings, thereby generating immersive force feedback and enhancing user experience in haptic interaction. For example, if the actuating element **400** vibrates at a particular frequency or rotates intermittently, users may feel a sensation similar to firing a machine gun. For example, if the actuating element **400** moves toward the trigger element **200**, the trigger may feel heavier to the user.

(29) In some embodiments, the force feedback module **100** further includes a control unit. Based on the signals generated by the sensing element **600**, the control unit may control the rotation, translation, or vibration of the actuating element **400** to control the amount and directions of generated driving force. In addition, in some embodiments, based on “the range of the angle of rotation of the trigger element **200**”, the control unit can define “the progress of applying force” given by a user.

(30) In particular, the angle between where the trigger element **200** does not start to rotate and where the trigger element **200** rotates to the limit may be referred to as “the range of the angle of rotation of the trigger element **200**”. In some embodiments, the range of the angle of rotation of the trigger element **200** is less than or equal to 80 degrees, such as about 15 degrees, about 20 degrees, about 25 degrees, about 30 degrees, about 35 degrees, about 40 degrees, about 45 degrees, or about 50 degrees, but it is not limited thereto. In addition, when the user does not start to press the trigger element **200**, “the progress of applying force” may be referred to as 0%. When the user feels that the press is done (i.e. the trigger element **200** rotates to the limit and cannot rotate further), “the progress of applying force” may be referred to as 100%.

(31) For example, if the range of the angle of rotation of the trigger element **200** is 50 degrees, for every degree the trigger element **200** rotates, the progress of applying force increases by 2%. If the range of the angle of rotation of the trigger element **200** is 25 degrees, for every degree the trigger element **200** rotates, the progress of applying force increases by 4%.

(32) When a user wants to use the force feedback module **100**, he can set the range of the angle of rotation of the trigger element **200** and the modes of force feedback in advance, according to different variables, including his hand size, personal preferences, the game he wants to play, etc. For example, if a user wants to simulate pulling the trigger of a gun, the user may set the range of the angle of rotation of the trigger element **200** to about 50 degrees; to have force feedback when the progress of applying force reaches 20%; and to have stronger force feedback when the progress of applying force reaches 100%.

(33) After the setting is completed, the control unit is able to calculate and control the initial position of the first transmission element **310**. Using the aforementioned settings, the angle between the trigger element **200** and the initial position of the first transmission element **310** is about 10 degrees. Therefore, when the progress of applying force is between 0% to 20%, no force feedback is generated. In addition, using to the aforementioned settings, the trigger element **200** would contact the first transmission element **310** when rotated 10 degrees, i.e. when the progress of applying force substantially reaches 20%, the sensing element **600** is able to sense that the first transmission element **310** is starting to rotate and generate signals to the control unit, and thus the control unit can control the actuating element **400** to generate a predetermined driving force. Using the aforementioned settings, when the progress of applying force reaches 100%, the control unit is

able to control the actuating element **400** to generate a greater force.

(34) The aforementioned settings are merely examples. Users may arbitrarily set the modes of force feedback. For example, a user may decide to have force feedback when starting to press, and the force feedback may increase as the progress of applying force increases. Under such settings, the trigger element **200** is in contact with the first transmission element **310** at the beginning, and the actuating element **400** may generate greater driving force as the trigger element **200** and the first transmission element **310** keep rotating. Alternatively, a user may decide to have force feedback for slow vibration when the progress of applying force reaches 50%, and to have force feedback for fast vibration when the progress of applying force reaches 80%. Under these settings, the vibration frequency of the actuating element **400** may increase. To sum up, the force feedback module **100** may adjust the initial position of the first transmission element **310** and the driving force generated by the actuating element **400**.

(35) It should be noted that, after every press is done by the user, the trigger element **200** may return to its original position by some elements that are able to generate elastic force for pulling the trigger element **200** back to its original position. In addition, the actuating element **400** may continuously generate the driving force, and the driving force may be transmitted to the first transmission element **310** via the transmission assembly **300**, so the first transmission element **310** may return to its initial position quickly. Therefore, every time a user applies force to the button or the trigger in the handheld controller, the force feedback module **100** can provide desirable modes of force feedback.

(36) As described above, a force feedback module is provided. The force feedback module may be provided in a handheld controller that includes a button or a trigger. The force feedback module includes a trigger element, an actuating element, and a transmission assembly. The trigger element may be disposed close to the fingers of a user. The actuating element may generate a driving force by rotation, translation, vibration, and the like. The driving force generated by the actuating element may be transmitted to the trigger element via the transmission assembly to provide immersive force feedback, thereby enhancing user experience in haptic interaction. In addition, when a user wants to use the force feedback module, he can set the modes of force feedback in advance, according to different variables, including his hand size, personal preferences, the game he wants to play, etc. For example, different force feedback can be provided based on the progress of applying force. Furthermore, after every press is done by the user, in addition to the trigger element, the transmission assembly may also return to its original position. Therefore, every time a user applies force to the button or the trigger in the handheld controller, the force feedback module can provide desirable modes of force feedback.

(37) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of this disclosure. Those skilled in the art should appreciate that they may readily use this disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of this disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of this disclosure. In addition, the scope of this disclosure is not limited to the specific embodiments described in the specification, and each claim constitutes a separate embodiment, and the combination of various claims and embodiments are within the scope of the disclosure.

Claims

1. A force feedback module comprising: a trigger element; an actuating element; and a transmission assembly disposed between the trigger element and the actuating element, wherein the transmission assembly comprises a first transmission element, and the first transmission element and the trigger

element change between a contact state and a non-contact state, wherein when the first transmission element and the trigger element are in the contact state, a driving force generated by the actuating element is transmitted to the trigger element via the transmission assembly to generate force feedback, and the first transmission element and the trigger element are configured to rotate about a common rotation axis until the first transmission element and the trigger element reach a rotational limit.

2. The force feedback module as claimed in claim 1, wherein the first transmission element comprises a protrusion facing the trigger element.

3. The force feedback module as claimed in claim 1, wherein the transmission assembly further comprises a second transmission element that meshes with the first transmission element.

4. The force feedback module as claimed in claim 3, wherein the transmission assembly further comprises a third transmission element and a transmission rod, and the second transmission element is connected to the third transmission element via the transmission rod.

5. The force feedback module as claimed in claim 4, wherein the transmission assembly further comprises a fourth transmission element that meshes with the third transmission element.

6. The force feedback module as claimed in claim 5, wherein the first transmission element comprises a linear gear, and the second transmission element, the third transmission element, and the fourth transmission element comprise a circular gear.

7. The force feedback module as claimed in claim 6, further comprising a pin, wherein the first transmission element comprises a transmission pivotally connected portion, and the pin passes through the transmission pivotally connected portion, the actuating element comprises a rotation shaft, the fourth transmission element is connected to the actuating element via the rotation shaft, when the first transmission element and the trigger element are in the contact state and the actuating element generates the driving force, the actuating element drives the fourth transmission element to rotate around the rotation shaft of the actuating element, the fourth transmission element drives the third transmission element and the second transmission element to rotate around the transmission rod, and the second transmission element drives the first transmission element to rotate around the pin.

8. The force feedback module as claimed in claim 1, further comprising a pin, wherein the trigger element comprises a trigger pivotally connected portion, the pin passes through the trigger pivotally connected portion, and when a force is applied to the trigger element, the trigger element rotates around the pin.

9. The force feedback module as claimed in claim 1, further comprising a sensing element sensing a position of the first transmission element and generates a signal.

10. The force feedback module as claimed in claim 9, further comprising a control unit controlling the amount and the direction of the driving force generated by the actuating element based on the signal.

11. The force feedback module as claimed in claim 9, wherein the transmission assembly further comprises a transmission rod embedded in the sensing element.

12. The force feedback module as claimed in claim 1, wherein the actuating element generates the driving force by rotation, translation, or vibration.

13. The force feedback module as claimed in claim 1, wherein the actuating element is a motor.

14. The force feedback module as claimed in claim 1, wherein the first transmission element comprises a transmission body that is L-shaped with a straight-line portion and a curved-line portion.

15. The force feedback module as claimed in claim 14, further comprising a pin around which the trigger element and the first transmission element rotate, wherein the transmission assembly further comprises a second transmission element that meshes with the first transmission element, and when the trigger element and the first transmission element reach the rotational limit and cannot rotate further, the second transmission element contacts the straight-line portion.

16. The force feedback module as claimed in claim 1, wherein a progress of applying force between 0% to 100% is defined by rotation range of the trigger element.

17. The force feedback module as claimed in claim 1, wherein when the first transmission element and the trigger element are in the non-contact state, a gap is formed between the first transmission element and the trigger element.
